Jesse Duku

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RESEARCH INTERESTS

Water resources · hydrologic and climate extremes · compound hazards · machine learning · remote sensing · computer vision

EDUCATION

Ph.D. Civil and Environmental Engineering — 2029 (expected)
University of California, Irvine — Hydrology & Water Resources
Advisor: Prof. Amir AghaKouchak

M.S. Civil and Environmental Engineering — Dec 2025
University of California, Irvine
Thesis: *The Shrinking Caspian Sea: Eco-hydrological Destabilization under Compound Human and Climate Stressors*

B.Sc. Civil Engineering — 2018 – 2022
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana
Thesis: *Punching Shear Prediction Using Support Vector Machines*

PUBLICATIONS

Peer-Reviewed Journal Articles

Duku, J., Tourian, M. J., Azarderakhsh, M., Abbasov, R., Mehran, A., Torabi Haghighi, A., Xenarios, S., Boschee, A., Babagiray, S., Sadegh, M., Shokri, N., Nazemi, A., Farahmand, A., Ashraf, S., Khujanazarov, T., Hassanzadeh, E., Najib, D., Placht, D., Mashtayeva, S., Rets, E., Norouzi, H., Madani, K., Shirzaei, M., Shafizadeh-Moghadam, H., Miao, C., Mirchi, A., Wang, S., & AghaKouchak, A. (2026). *The shrinking Caspian Sea: Eco-hydrological responses to human and climate pressures.* Earth's Future, 14(6), e2025EF008028.

AghaKouchak, A., Hjelmstad, A., Lucy, J., **Duku, J.**, Sadhasivam, N., Werth, S., Babagiray, S., de Oliveira, D. Y., Boschee, A., Sadegh, M., Vahedifard, F., Abatzoglou, J. T., Turco, M., Ortiz, M. S., Shirzaei, M., Azimi, P., Ferguson, L., & Allen, J. G. (2025). *Building urban fire resilience to enhance national security.* Nature Cities, 2(9), 778–780.

Boschee, A., Alborzi, A., Alexander, A. C., Arheimer, B., Bel Hadj Ali, S., … **Duku, J.**, … & AghaKouchak, A. (2026). *The Coupled Hydrology–Human Activity Information (CHHAI) dataset: a global benchmark dataset for model comparison and evaluation.* Hydrological Sciences Journal (in press).

Conference Presentations

Babagiray, S., AghaKouchak, A., Hjelmstad, A., **Duku, J.**, Boschee, A., et al. (2025). *Assessing compound hazards: Integrating fire and post-fire debris flow risk in Southern California.* American Geophysical Union (AGU) Annual Meeting, New Orleans, LA.

RESEARCH EXPERIENCE

Graduate Student Researcher — Jun 2024 – Present
University of California, Irvine — Advisor: Prof. Amir AghaKouchak
- Research hydrologic and climate extremes, compound hazards, and water resources using statistical hydrology and machine learning.
- Process and analyze satellite remote-sensing observations and hydrological reanalysis products to characterize change across large basins.

Research Assistant — Nov 2022 – Feb 2023
Regional Water and Environmental Sanitation Centre (RWESCK), KNUST
- Conducted laboratory experiments and prepared technical reports.
- Summarized survey data collected through structured field interviews.

SELECTED RESEARCH PROJECTS

Global Food Security under Compound Climate Shocks 2026 – Present

- Using climate data, AI, and network science to model how compound climate shocks propagate through the international food system and to predict acute food insecurity outcomes (IPC Phase 3+).

Composite Flood Distribution Jan 2025 – Present

- Applied probability distributions to model composite flood frequency and magnitude, and analyzed the joint probability of multiple flood drivers to improve risk assessment.

Caspian Sea Hydrological Dynamics & Climatic Influences Jun 2024 – Dec 2024

- Quantified hydrological change and climatic drivers behind declining Caspian Sea levels; developed water-balance projections to inform transboundary water policy. (Basis of the Earth's Future, 2026 article.)

Water-Quality Monitoring for Artisanal Small-Scale Mining (ML) Jan 2023 – Oct 2023

- Developed empirical models integrating multi-sensor data to monitor water quality in mining-affected catchments.

Plastic-Free Rivers Hackathon Jul 2023 – Aug 2023

REVA University

- Built a YOLOv8 computer-vision model to detect plastic in rivers; deployed on Hugging Face.

TEACHING EXPERIENCE

Teaching Assistant — CEE 81B: Civil Engineering Practicum II Winter 2026

University of California, Irvine

- Supported instruction in surveying, GIS fundamentals, and engineering visualization; led lab sessions and assisted students with assignments.

Teaching Assistant — Department of Civil Engineering Sep 2022 – Dec 2022

Kwame Nkrumah University of Science and Technology (KNUST)

- Taught tutorials in water resource engineering and AutoCAD, facilitating sessions for 100+ students per semester.

PROFESSIONAL EXPERIENCE

Civil Engineering Trainee Sep 2021 – Dec 2021

Golden Mainland Company Limited

- Conducted site surveys, location markings, and construction monitoring across multiple projects.
- Performed field investigations to gather data on structural performance.

TECHNICAL SKILLS

Programming & ML: Python (data analysis, machine learning), computer vision

GIS & Remote Sensing: QGIS, multi-sensor and satellite data processing

CAD: AutoCAD, Revit, Civil 3D, SSA

Other: Laboratory methods; Microsoft Office (Excel, PowerPoint, Word)

CERTIFICATES & TRAINING

- Aspire Institute Leadership Program.
- Data Science with Python — certificate.

CONFERENCES

- American Geophysical Union (AGU) Annual Meeting — presenter, 2025.
- Sustainable Infrastructure Conference, Ghana Consulting Engineers Association (GCEA), Accra, Ghana, 2023.

PROFESSIONAL MEMBERSHIPS

American Geophysical Union (AGU) · American Society of Civil Engineers (ASCE) · National Society of Black Engineers (NSBE) · Black in AI